

JEONG GYU (BRIAN) LEE

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EDUCATION

Master of Science, Electrical Engineering, California State University, Long Beach
GPA: 4.0

December 2026

Bachelor of Mathematics, University of California, Santa Barbara
GPA: 3.69

June 2020

TECHNICAL SKILLS

Experimental Characterization: UV-Vis-NIR spectroscopy, Raman spectroscopy, SEM/FIB, spectrum analyzer, vector network analyzer, DC power supply, digital multimeter

Simulation and Analysis: Lumerical FDTD, MATLAB, Keysight ADS

Hardware and Prototyping: RF prototyping, mixed-signal circuits, Arduino, optical circuits

RESEARCH EXPERIENCE

Research Assistant

November 2024 – Present

California State University, Long Beach

- Designed and simulated plasmonic devices in Lumerical FDTD, achieving a 25% improvement in optical transmission and $\times 26$ electric-field enhancement relative to baseline designs using Raman spectroscopy.
- Built, aligned, calibrated, and maintained a broadband spectroscopy and microscope-imaging setup for photonic-device characterization.
- Developed MATLAB workflows to automate spectroscopy processing and microscopic-image analysis, including extraction of quantitative inspection metrics for 3D-printed structures.
- Mentored and trained two student researchers in experimental procedures, equipment operation, and data analysis while coordinating assignments and reviewing technical results.
- Characterized surface morphology and multi-layered structures with SEM/FIB and EDS.

ADDITIONAL EXPERIENCE

Audio/Visual Systems Integrator

March 2020 – Present

Independent Contractor

- Designed, integrated, tested, and troubleshoot client-specific audio/visual systems by translating operational requirements into hardware configurations while meeting budget and schedule constraints.

PUBLICATIONS

- **J. G. Lee, Y. Trivedi, A. Ahmed*** “Free Space to MIM Slit Plasmonic Waveguide Coupling via a Quarter-Wavelength Impedance Transformer” (2026)
- **J. G. Lee, K. Luhm, S. Carrillo, A. Ahmed, A. Lee*** “Review of Light Detection and Ranging (LiDAR) Systems from Rotating Mirrors to Chip-Scale Beam Steering” (2026)

PROJECTS

2.5 GHz to 5 GHz Frequency Doubler

- Designed and simulated a BFP420 BJT frequency doubler in Keysight ADS using harmonic-balance analysis, producing a 5 GHz output of -7 dBm from a 2.5 GHz input while suppressing undesired harmonics to -59 dBm.
- Fabricated an FR-4 prototype with copper-tape RF traces and characterized its frequency response using a spectrum analyzer and VNA.

Acoustic Piano to Digital MIDI Controller

- Designed a mixed-signal sensing interface that converted mechanical key motion into digital time-of-flight measurements using TCST2103 optical encoders, 555-timer Schmitt-trigger circuits, and an Arduino Nano.

9 GHz Power Amplifier

- Designed and simulated matching networks for a single-stage power amplifier achieving 12 dB gain within a 10% bandwidth from 9 GHz using an NE67383 MESFET in Keysight ADS.

HONORS

- Best Poster Award, IEEE Silicon Beach Annual Conference November 2025
- Ostrow-Bruckner Mathematics Excellence Scholarship October 2018

CONFERENCES ATTENDED

- **J. G. Lee**, Y. Trivedi, A. Lee, A. Ahmed, ‘Enhanced Optical Transmission via Nanoslit Transformer Design’. 2026 MRS Spring Meeting. 04/27/2026, Honolulu, HI
- P. Du, **B. Lee**, H. Nguyen, A. Ahmed, A. Lee*, “Thermo-Decoupled Photon-Stimulated Enhancement in Prussian Blue Electrodes for Aqueous Sodium-Ion Batteries” 2026 MRS Spring Meeting. 04/27/2026, Honolulu, HI
- S. Carrillo, **J. G. Lee**, Y. Wang, K. Ratzel, V. Xia, Y. Suganuma, A. Ahmed, A.-A. Dhirani, A. Lee*, “Effects of TiO₂ Cavity Thicknesses on Resonance Excitation Wavelengths of Plasmonic Structures” 2026 MRS Spring Meeting. 04/27/2026, Honolulu, HI
- S. Carrillo, **J. G. Lee**, Y. Wang, K. Ratzel, V. Xia, Y. Suganuma, A. Ahmed, A. Dhirani, A. Lee, ‘Broadband and Large-Area Field Enhancement for SERS via Hybrid Plasmonic-Dielectric Cavity Coupling’. 38th CSULB Student Research Competition. 2/27/2026, Long Beach, CA
- **J. G. Lee**, S. Carrillo, Y. Wang, K. Ratzel, V. Xia, Y. Suganuma, A. Ahmed, A. Dhirani, A. Lee, ‘Broadband and Large-Area Field Enhancement for SERS via Hybrid Plasmonic-Dielectric Cavity Coupling’. 2026 Engineering Horizons. 2/20/2026, Long Beach, CA
- **J. G. Lee**, S. Carrillo, Y. Wang, K. Ratzel, V. Xia, Y. Suganuma, A. Ahmed, A. Dhirani, A. Lee, ‘Broadband and Large-Area Field Enhancement for SERS via Hybrid Plasmonic-Dielectric Cavity Coupling’. 2025 MRS Fall Meeting. 12/4/2025, Boston, MA
- **J. G. Lee**, Y. Trivedi, A. Lee, A. Ahmed, ‘Enhanced Optical Transmission in Multilayer Nanoslit Structures – Theory and Simulation’. IEEE Green Energy and Smart Systems Conference 2025. 11/3/2025, Long Beach, CA
- P. Du, **J. G. Lee**, H. Nguyen, A. Ahmed, A. Lee, ‘Thermo-Decoupled Photon-Stimulated Enhancement in Prussian Blue Electrodes for Aqueous Sodium-Ion Batteries’. IEEE Green Energy and Smart Systems Conference 2025. 11/3/2025, Long Beach, CA
- **J. G. Lee**, Y. Trivedi, A. Lee, A. Ahmed, ‘Enhanced Optical Transmission in Multilayer Nanoslit Structures – Theory and Simulation’. IEEE Silicon Beach Annual Conference (ISBAC 2025). 11/3/2025, Long Beach, CA
Best Poster Award